CODEFEST 2.

Given:

Peak Compute = 10 FLOPS (FP32)

Peak DRAM BW = 320 GB/s.

ridge point = $\frac{10,000}{320} \approx 31.25$ FLOP/byte

→ Kernel A: A, B, C matrices GEMM

$N = 1024$

memory = FP32 = 4 bytes.

FLOPS = $2 \times N^3 = 2 \times (1024)^3$

$= 2.147 \times 10^9$

bytes transferred = $3N^2 \times 4 = 3 \times (1024)^2 \times 4$

$= 1.258 \times 10^7$ bytes.

$I = \frac{N}{6} = \frac{1024}{6} \approx 170.67$ FLOP/byte.

→ Kernel B: Vector Addition.

length = 4,194,304 elements

memory = FP32 = 4 bytes.

FLOPS = N = 4,194,304

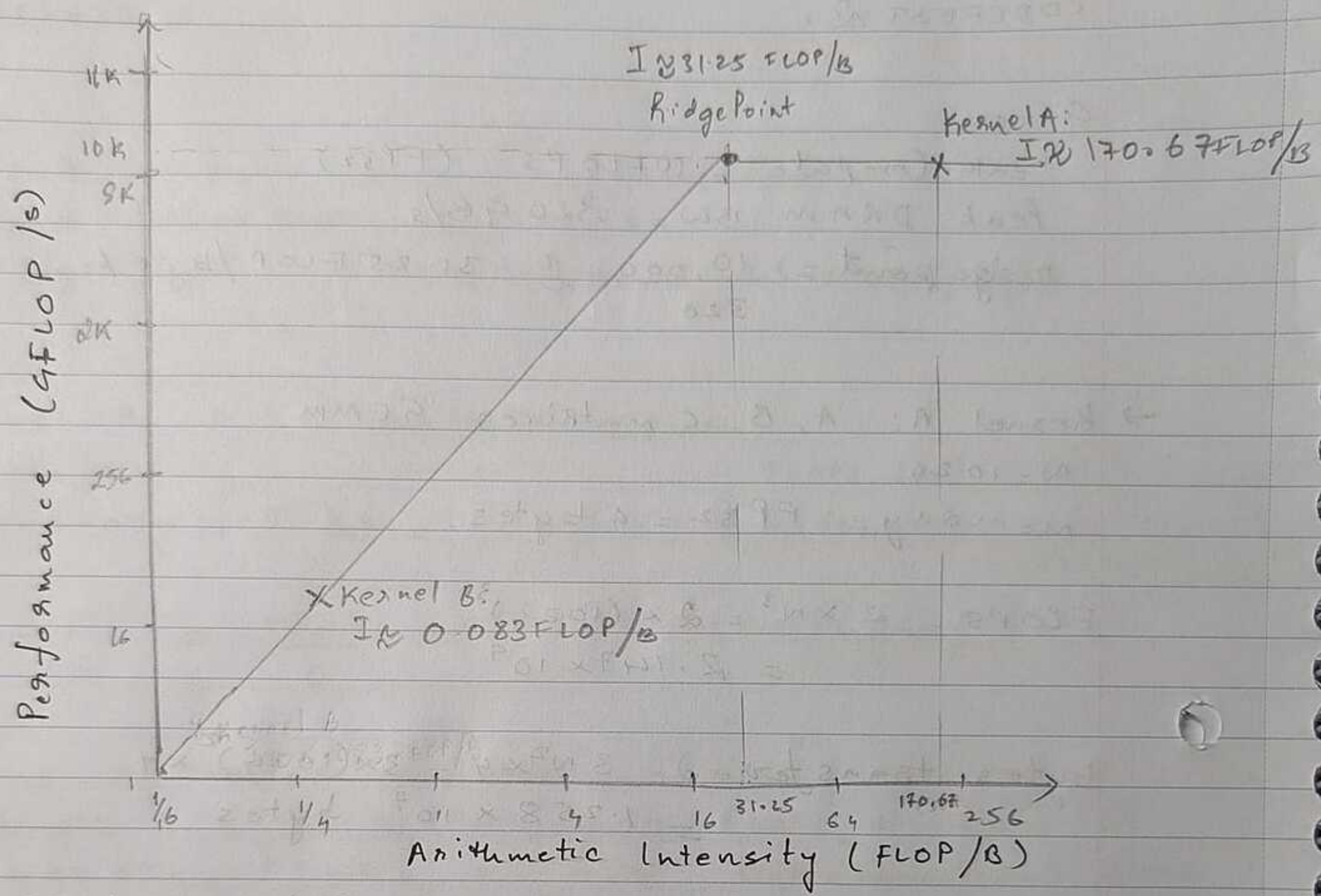
Bytes transferred = $3N \times 4$

$= 3 \times 4,194,304 \times 4$

$= 5.033 \times 10^7$ bytes

$I \approx 0.083$ FLOP/byte.

This number is constant for vector addition.



ROOFLINE MODEL

4) a) Kernel A: Compute Bound.

b) - Attainable performance: 10K GFLOP/s.

c) - Currently, Kernel A is on the compute ceiling of the roofline model for this hardware. This implies data is being fed fast enough but processor cannot perform faster execution on this.

- Architectural change to improve performance is to use lower precision, that is, instead of using FP32 we can compute using FP16. This change doubles the throughput. Because the multiple data that can be accessed doubles, thus, doubling the number of operations per cycle.

4 a) Kernel B : Memory Bound

b) - Attainable Performance = Peak BW × Arithmetic Intensity.

$= 320^{GB/s} \times 0.083$ FLOP/byte.

$= 320 \times 10^{10} \times 0.083$

$= 26.56$ GFLOP/s.

c) - Kernel B, is on the memory bound slope. This implies it is moving more data than ~~performing~~ executing / computing.

- Architectural change to improve performance is combining / fusing previous or next instructions with the vector operations. This basically increases the compute capability for the same amount of data movement.